Virtual University of Pakistan

BIO204

Principles of Biochemical Engineering

Midterm past papers solved

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Created By

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Bio204 26 dec 2018

Steady state nd quasi stady state difference, effect of y mue and ks 5
Requirements for fermenter design 5

Ans: There is no universal bioreactor. The general requirements of the bioreactor are as follows:
1. The vessel should be capable of being operated aseptically for a number of days and should be reliable in long-term operation and meet the requirements of containment regulations.
2. Adequate aeration and agitation should be provided to meet the metabolic requirements of the micro-organism. However, the mixing should not cause damage to the organism.
3. Power consumption should be as low as possible.
4. A system of temperature control should be provided.
5. A system of pH control should be provided.
6. Sampling facilities should be provided.
7. Evaporation losses from the fermenter should not be excessive.
8. The vessel should be designed to require the minimal use of labour in operation, harvesting, cleaning and maintenance.
9. Ideally the vessel should be suitable for a range of processes, but this may be restricted because of containment regulations.
10.The vessel should be constructed to ensure smooth internal surfaces, using welds instead of flange joints whenever possible.
11.The vessel should be of similar geometry to both smaller and larger vessels in the pilot plant or plant to facilitate scale-up.
12.The cheapest materials which enable satisfactory results to be achieved should be used.
13.There should be adequate service provisions for individual plants.

Substate nd bio yeild relation 2

Ans: The **Growth Yield** (Y) is the increase in cell mass (dX) which results due to consumption of a small amount of substrate (dS)
Y=dX/dS

Qp nd mue relation 3

Ans: Product concentration in a fed batch system over the time of the fermentation will be dependant on the relationship between qp and u. If qp is growth related than it will change as u thus, the product concentration remains constant. If qp is constant and independent of uthan product concentration will start of the cycle when Dp is greater $q_p x$ but will rise with time as D decreases and $q_p x$ become greater than Dp.

Aggelator function 3

Ans: Agitation plays an important mixing and shearing role in fermentation processes. It not only improves mass and oxygen transfer between the different phases, but also maintains homogeneous chemical and physical conditions in the medium by continuous mixing.

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Question No1
Describe relationship between qp and u.(3)

Ans: Product concentration in a fed batch system over the time of the fermentation will be dependant on the relationship between qp and u. If qp is growth related than it will change as u thus, the product concentration remains constant. If qp is constant and independent of uthan product concentration will start of the cycle when Dp is greater $q_p x$ but will rise with time as D decreases and $q_p x$ become greater than Dp.

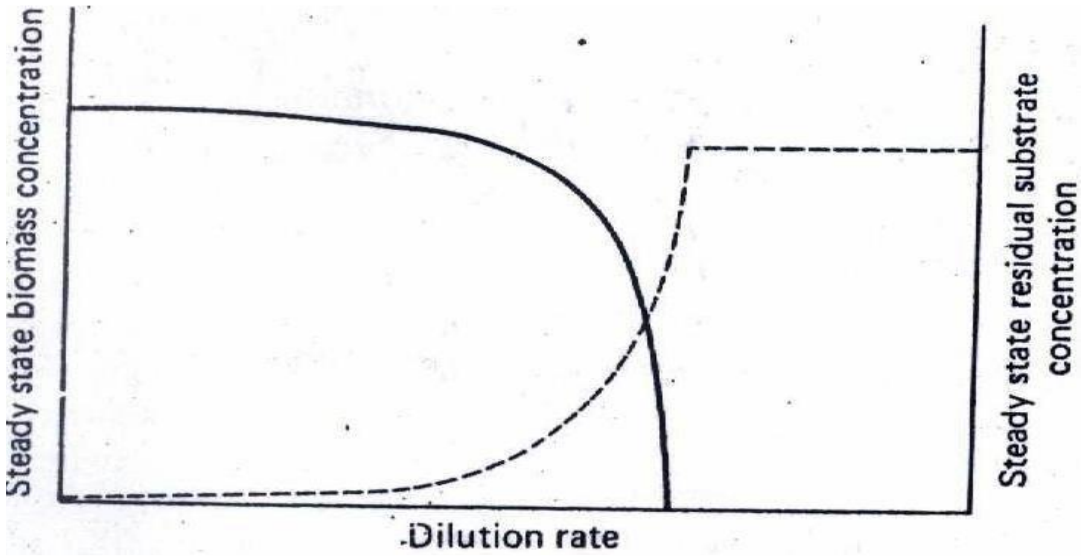
Question No2
Write two functions of agitation.(3)

Ans: Agitation plays an important mixing and shearing role in fermentation processes. It not only improves mass and oxygen transfer between the different phases, but also maintains homogeneous chemical and physical conditions in the medium by continuous mixing.

QuestionNo3
What is the major difference between quasi steady state and steady state of chemostat of fed-batch culture also write effects of values of Y, Umax and Ks.(5)

Ans: The major difference between the quasi steady state and steady state of chemostat of a fed batch culture is that μ is constant in the chemostate but decreases in the fed batch culture.

The effect of dillution rate on the steady state biomass and residual substrate concentrations in a chemostate of a microorganisms with a low ks value for the limiting substrate compared with the initial concentration.



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With increasing dilution, the residual substrate concentration increases only slightly until D approaches umax when s increases significantly. The dilution rate at which x equal zero is termed the critical dilution rate $\left(D_{crit}\right)$ and is given by the equation:

$$D_{crit} = \mu_{\max} S_R \backslash \left(K_S + S_R\right)$$

The kinetic characteristics of the organism are discribed by the numerical values of the constants y, umax and ks.

The value of Y affects the steady state biomass concentration, The value of umax affects the maximum dilution rate that may be employed and the value of Ks affects the residual substrate concentration and also the maximum dilution rate that may be used.

The continuous culture behavior of a hypothetical bacterium with high *Ks* for the limiting substrate compared with the initial limiting substrate concentration. With increasing dilution rate, the residual substrate concentration increases significantly to support the increased growth rate. Thus, there is a gradual increase in *s* and a decrease in *x* as *D* approaches *Dcrit*. Thus, *Dcrit* is affected by the constants, *μmax* and *Ks*, and the variable, *SR;* the larger *SR* the closer is *μmax*. However, *μmax* cannot be achieved in a simple steady state chemostat because substrate limited conditions must always prevail.

Question No 4
What is Agitators radial flow and axial flow characteristics. (5)

Ans:

- With radial flow mixing, the liquid flow from the impeller is initially directed towards the wall of the reactor; i.e. along the radius of the tank.
- □ With axial flow mixing, the liquid flow from the impeller is directed downwards towards the base of the reactor, i.e. in the direction of the axis of the tank.
- □Radial flow impellers are primarily used for gas-liquid contacting (such as in the mixing of sparged bioreactors) and blending processes.
- □ Axial flow impellers provide more gentle but efficient mixing and are used for reactions involving shear sensitive cells and particles.
- Radial flow mixing is not as efficient as axial flow mixing.
- For radial flow impellers, a much higher input of energy is required to generate a given level of flow.
- Radial flow impellers do and are designed to, generate high shear conditions. This is achieved by the formation of vortices in the wake of the impeller.

Question No 5
On the of productivity the biological products on factore of biomass? 2

Ans: In case of Product:
Sometimes □x□ is itself product (e.g. Baker's Yeast) and sometimes product is directly linked/associated with biomass (e.g. DNA , Biomass, Primary metabolites).

Question No 6
What is chemostate(2)

Ans: A chemostat is a nutrient-limited self balancing culture system which may be maintained in a steady state over a wide range of sub-maximum specific growth rates.

Bio 204
25 June 2018, 12:00 pm

Total questions 26
20 mcq's + 6 questions

1)Describe relationship between qp and u.

Ans: Product concentration in a fed batch system over the time of the fermentation will be dependant on the relationship between qp and u. If qp is growth related than it will change as u thus, the product concentration remains constant. If qp is constant and independent of uthan product concentration will start of the cycle when Dp is greater $q_p x$ but will rise with time as D decreases and $q_p x$ become greater than Dp.

2)Why use of glass is preferable in fermenter?

Ans: On a small scale (1 to 30 L) it is possible to use glass and/or stainless steel. Glass is useful because it gives smooth surfaces, is non-toxic, corrosion proof and it is usually easy to examine the interior of the vessel.

3)Write two functions of agitation.

Ans: Agitation plays an important mixing and shearing role in fermentation processes. It not only improves mass and oxygen transfer between the different phases, but also maintains homogeneous chemical and physical conditions in the medium by continuous mixing.

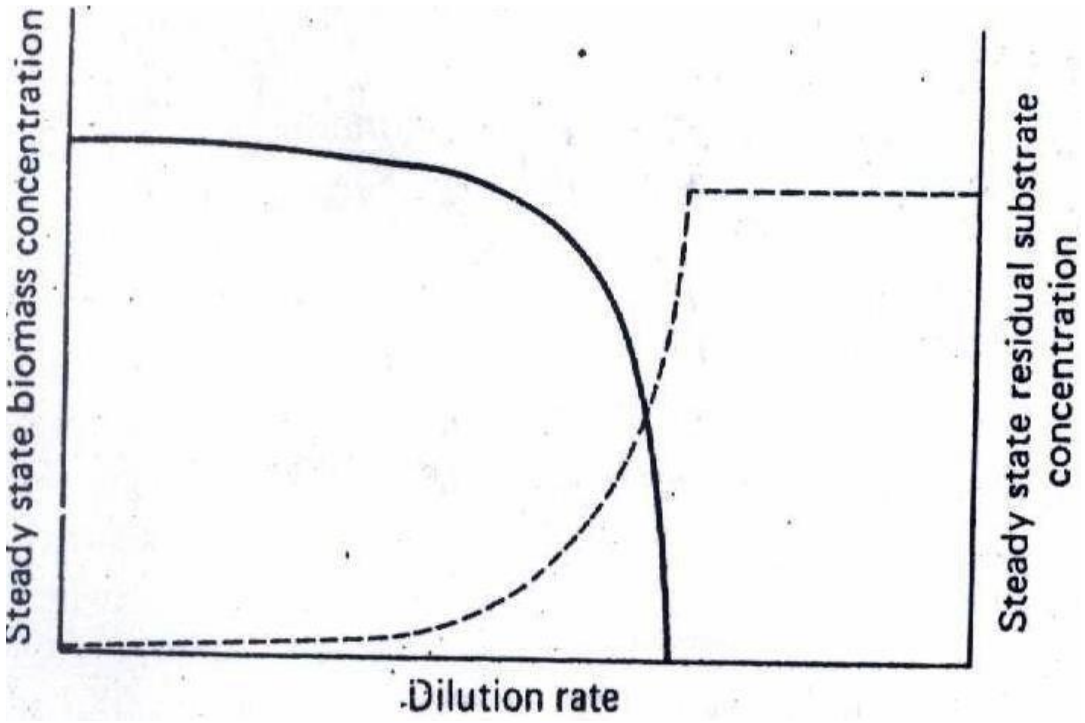
4)Define quasi steady state.

Ans: The total biomass in the culture increase with time, cell concentration remains virtually constant that is $(dx/dt) \approx 0$ and therefore $\mu \approx D$ it is called quasi steady state.

5)What is the major difference between quasi steady state and steady state of chemostat of fed-batch culture also write effects of values of Y, Umax and Ks.

Ans: The major difference between the quasi steady state and steady state of chemostat of a fed batch culture is that μ is constant in the chemostate but decreases in the fed batch culture.

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6)Differentiate between axial and radial flow.

Ans: With radial flow mixing, the liquid flow from the impeller is initially directed towards the wall of the reactor; i.e. along the radius of the tank.
 With axial flow mixing, the liquid flow from the impeller is directed downwards towards the base of the reactor, i.e. in the direction of the axis of the tank.
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----------Good Luck----------